DIRECT FUEL INJECTION SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Outline

- In the direct fuel injection system, the temperature in the combustion chamber is lowered by injecting fuel directly into the combustion chamber using multi-hole injectors and atomizing the fuel. In addition, improved engine output and fuel economy have been realized by increasing the air density during compression.
- High response has been realized because there is no time lag from when the fuel injection starts until the fuel is provided to the combustion chamber.

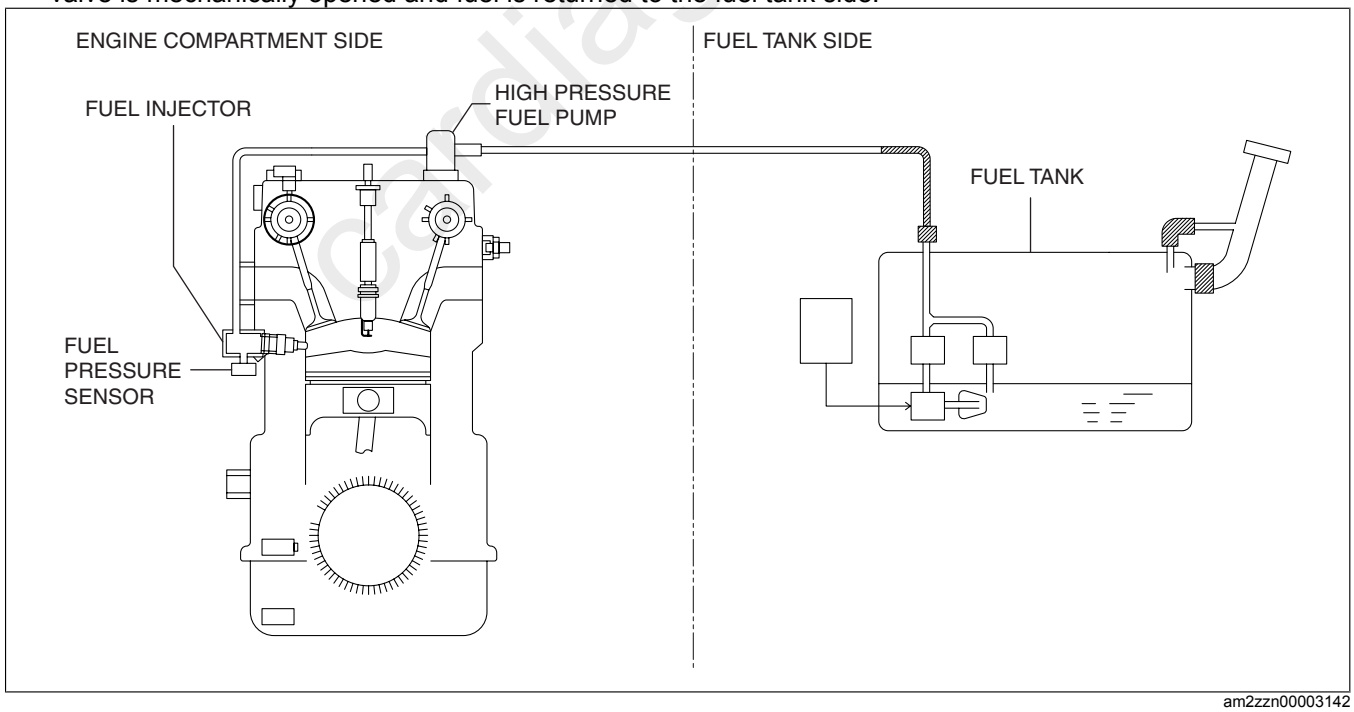
Structure

- The direct fuel injection system consists of the following parts.

Part name	Reference
High pressure fuel pump	(See HIGH PRESSURE FUEL PUMP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Fuel injector	(See FUEL INJECTOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Fuel distributor	(See FUEL DISTRIBUTOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Fuel pressure sensor	(See FUEL PRESSURE SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Relief valve	—

Operation

- Fuel from the high pressure fuel pump injects fuel directly into the cylinder using the fuel injector via the fuel distributor. Injected, atomized fuel during the intake stroke is diffused uniformly in the combustion chamber by the flow of intake air, achieving a uniform air-fuel mixture. For the fuel injection control, refer to the CONTROL SYSTEM, FUEL INJECTION CONTROL. (See FUEL INJECTION CONTROL SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
- The fuel pressure sensor detects the fuel pressure in the fuel distributor and sends a fuel pressure signal to the PCM. Based on the signal from the fuel pressure sensor, the PCM controls the high pressure fuel pump so that the fuel pressure is in accordance with the driving conditions. For the high pressure fuel pump control, refer to CONTROL SYSTEM. (See HIGH PRESSURE FUEL PUMP CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
- Fuel pressure is adjusted by the spill valve control solenoid valve, and a relief valve is equipped to ensure safety in case of a malfunction. If the fuel pressure in the high pressure fuel pump reaches the set pressure, the relief valve is mechanically opened and fuel is returned to the fuel tank side.



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